

ERAY KULKIZAGA

Software Engineer | Python Backend & AI Systems

+90 539 603 39 41 | Türkiye

eraykulkizaga@hotmail.com | [linkedin.com/in/eraykulkizaga](https://www.linkedin.com/in/eraykulkizaga) | www.eraykulkizaga.com | github.com/ErayKulkizaga

Software Engineer specializing in Python backend and AI systems. Builds production-oriented APIs and retrieval-augmented applications using FastAPI, PostgreSQL, Redis, Celery, Docker and LLM APIs. Experienced in real-time inference, computer vision and deploying measurable AI services.

SKILLS

Languages: Python, SQL, TypeScript

Backend: FastAPI, REST, WebSocket, Pydantic, SQLAlchemy, Alembic

Data & Async: PostgreSQL, pgvector, Redis, Celery

AI Systems: LLM APIs, RAG, ONNX Runtime, OpenCV, TensorFlow

Testing & DevOps: pytest, Ruff, Pyright, Docker, GitHub Actions

EDUCATION

Eastern Mediterranean University — Famagusta, Northern Cyprus

Sep 2021 – Jun 2026

B.Sc. in Software Engineering

- ABET-accredited | 100% English-taught Program
- 2 Times Dean's High Honor List Student | Final Semester GPA: 3.83/4.00
- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Data Science, Linear Algebra, Probability

SELECTED PROJECTS

SignTurk | Real-Time Turkish Sign Language Recognition System | [GitHub](#)

Nov 2025 – Jun 2026

Tech: Python, FastAPI, WebSocket, ONNX Runtime, RTMPose, TensorFlow, OpenCV

- Built an end-to-end recognition system for 226 Turkish Sign Language classes, processing more than 35,000 AUTSL videos through a custom landmark-extraction and sequence-classification pipeline.
- Developed an ONNX-based pose and hand-landmark pipeline that transformed camera frames into structured temporal features for model training and real-time inference.
- Engineered a FastAPI/WebSocket service handling live video ingestion, sequence buffering, inference, confidence scoring and latency diagnostics.
- Achieved 94.17% Top-1 accuracy, 99.49% Top-5 accuracy and 94.00% macro-F1; received the Outstanding Project Award among 30 graduation projects.

PROFESSIONAL EXPERIENCE

Microsoft Türkiye | AI Innovators Summer Internship Participant

Aug 2026 – Sep 2026 (Ongoing)

- Completed a mentorship-based Microsoft Türkiye program combining weekly technical sessions with an independently delivered AI capstone.
- Built a local retrieval-augmented generation application with Microsoft Foundry Local, implementing document ingestion, context retrieval, grounded response generation, and local model execution.
- Published the source code and technical documentation on GitHub and delivered a final technical presentation for program completion.

FlyRank.ai | Backend AI Engineer Intern

Jun 2026 – Aug 2026 (Ongoing)

- Completed an accelerated Backend AI Engineering internship focused on automation, data-intensive applications, agent workflows, documentation, and token/cost-aware AI system design.
- Built and documented production-oriented backend systems spanning REST/CRUD APIs, database integration, Docker, authentication, AI API integration, background jobs, responsible scraping, and PDF report generation.
- Applied AI fluency practices to prompt engineering, workflow automation, agent development, evaluation, security hardening, capstone delivery, and technical case-study communication.

Iskur Holding | Enterprise Data Software Engineering Intern

Jun 2024 – Aug 2024

- Optimized SQL-based enterprise data retrieval workflows using SAP Open SQL joins, reducing processing time by 41%.
- Developed backend validation and automation logic for contract, barcode, inventory, and operational cost workflows.
- Implemented transactional logging and structured audit trails to improve system monitoring, reporting, and traceability.

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — Stanford University / DeepLearning.AI | [Credential](#)

Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI | [Credential](#)

Tools for Data Science — IBM | [Credential](#)

LANGUAGES

English | C1 Advanced

Turkish | Native